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Educational Radio Frequency Transmitter

Group: 2016.043

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11 June 2015

This is an excellent report, scoring 49.5 / 50

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1. Introduction

1.1. Motivation

In the past year, the number of mobile devices connected globally grew to 7.4 billion from 6.9 billion [1]. With the prevalence of wireless technology, it is essential to educate future engineers about the fundamentals of wireless systems and how they function. While most universities have laboratory exercises using MATLAB Simulink, only some can offer students hands on experience with cellular hardware. Current educational Radio Frequency (RF) and microwave equipment is expensive and performs poorly with multiple transmitting frequencies.

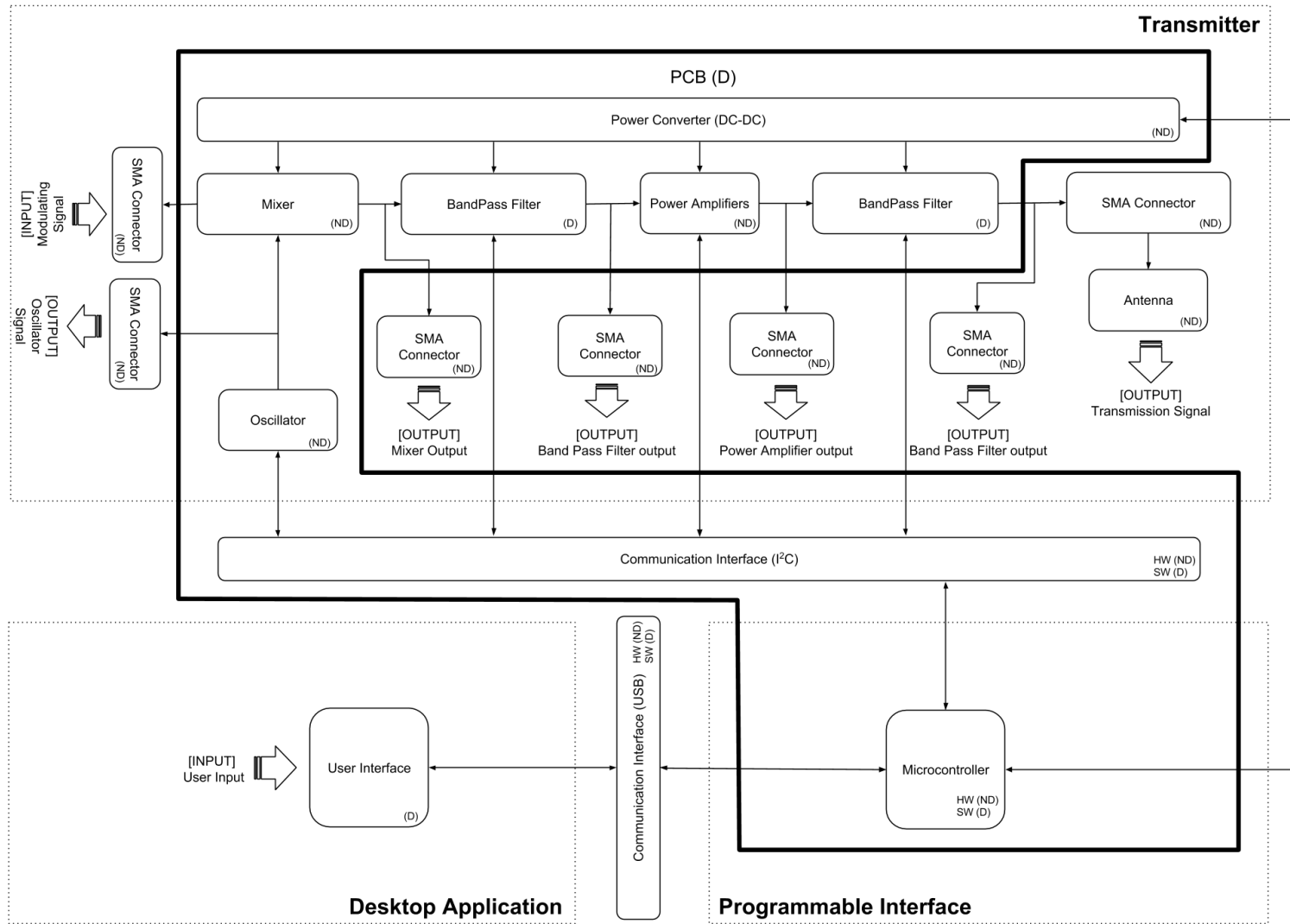
1.2. Project Objective

The objective of this project is to design a reconfigurable RF transmitter module. The module allows students to adjust the transmitting frequency and the power amplifier gain. Students can measure signal characteristics, such as noise figure, gain, and distortion at different stages of the transmitter.

1.3. Block Diagram

The high level design of the configurable transmission module is described in Figure 1. The system can be divided into 3 major subsystems: a transmitter, a programmable interface and a desktop application. The transmitter module is a device that outputs a cellular radio signal. The programmable interface is a system consisting of a microcontroller and computer. The interface controls the transmitter settings. The desktop application is used to adjust the transmitter settings. There are two communication interfaces between the subsystems. The first communication interface is used by the desktop application to communicate with the microcontroller. The other communication interface is used between the microcontroller module and the transmitter module. The system accepts two types of inputs: user defined transmitter properties and a modulating signal. The system also has six outputs, which are located at each connector at each transmitter stage.

The transmitter can be divided into seven building blocks, excluding connectors. Each of these blocks represents a component required to build a cellular transmitter [2]. The power converter, power amplifier, mixer and oscillator are subsystems that have off-the-shelf electronic components available. These components only require routine design in order to be integrated into the system. The seven connectors and the antenna do not require any design. Store bought bandpass filters do not have the functionalities that are required for the project. In order to meet the design specifications, an adjustable bandpass filter design operating at high frequencies is needed. A printed circuit board (PCB) design is needed to place and route the components. Within the PCB design, matching networks between the subsystems need to be optimized in order to minimize voltage reflections. The microcontroller can be purchased from a vender and no hardware design is required. However, software design is needed for the firmware of the microcontroller in order to control the transmitter. The desktop application is also a simple subsystem with a user interface where software design is needed. Finally, the communication interface uses standard bus communication protocols, which does not require any hardware design.



(D) = the subsystem will be designed as part of the project
 (ND) = the subsystem will not be designed as part of the project
 (HW) = the hardware component of the subsystem if applicable
 (SW) = the software component of the subsystem applicable

Figure 1 - Block Diagram of the Educational Radio Frequency Transmitter

2. Project Specifications

2.1. Functional Specifications

The functional specifications of the Educational Radio Frequency Transmitter are shown in Table 1.

Table 1. Functional specifications of the Educational Radio Frequency Transmitter

Specification	Description	Classification
Power consumption	The maximum total power consumption of the transmitter module is less than 4.5 W when the input supply voltage is 5 V or 9 W when the input supply voltage is 10 V. This power consumption constraint applies to operation of the transmitter module in standby and transmission mode. Standby mode is when the transmitter module output power is less than 5 mW. Transmission mode is when the module RF output power is greater than 5 mW.	Essential
RF output power	The RF output power of the transmitter module is below the maximum output power of 100 mW when transmitting at all configurable carrier frequency values.	Essential
Amplitude modulation (AM) and amplitude shift keying (ASK) support	The output of the transmitter module is an amplitude modulated signal centered at the configured carrier frequency.	Essential
Configurable power amplifier gain	In the application interface, the user is able to set the gain within the allowable range of the power amplifier stage. The allowable gain range is 10 dB to 20 dB.	Essential
Configurable carrier frequency	The user is able to set the carrier frequency and gain of the transmitted signal through the application interface. The carrier frequency range of the transmitted signal is limited to 800 MHz to 1.9 GHz. The carrier frequency that the user chooses is limited to the capabilities of the hardware system.	Essential
Maximum carrier frequency deviation	The maximum carrier frequency deviation from the configured carrier frequency is 1%. The range of configurable frequencies is between 800 MHz and 1.9 GHz. For example, for a 1 GHz signal, the maximum frequency deviation is ± 10 MHz.	Essential
Configurable bandpass filter stages	The bandpass filter stage is adjustable to allow the transmission signals around the user	Essential

	configured carrier frequencies. The allowable carrier frequencies ranges are between 800 MHz and 1.9 GHz.	
Overcurrent protection at the input of the transmitter for the message signal	The maximum current of the input signal that the transmitter can draw before triggering over current protection is 500 mA. Over current protection enables the isolation of the input from the rest of the transmitter module to prevent damage to other components on the board.	Essential
Overvoltage protection at the input of the transmitter for the message signal	The maximum input voltage across the input terminals before triggering overvoltage protection is 110% of the input supply voltage. Overvoltage protection enables the isolation of the input from the rest of the transmitter module to prevent damage to other components on the board.	Essential
Overcurrent protection at the input supply	When the input supply current drawn from the supply goes beyond 1 A, overcurrent protection should be triggered. This prevents the damage to internal components on the board.	Essential
Overvoltage protection at the input supply	When the input supply voltage goes beyond 110% of the specified supply voltage, overvoltage protection should be triggered. This prevents the damage to internal components on the board.	Essential
Quadrature amplitude modulation (QAM) support	Through the use of an in-phase and quadrature modulator, the transmitter is able to perform QAM. The output of the transmitter is a QAM signal centered on the configured carrier frequency.	Non-essential
Configurable modulation settings	The user is able to set the modulation scheme used in the transmitter. The allowable modulation schemes are AM, ASK, and QAM.	Non-essential

2.2. Non-functional Specifications

In addition to the functional specifications, the project also contains several non-functional specifications that are outlined in Table 2. The case of the module is an essential non-functional specification because it protects the circuit of the module from being tampered by students, but still allows students to see how each component of the circuit is connected. The connectors in the module are essential specifications because they allow user to perform characterization at each stage of the transmitter module. Characterization includes measuring the noise figure, the gain, and distortion of each stage of the transmitter. There should be a total of seven 50 Ω terminated Sub-Miniature version A (SMA) connectors. The cost of the module prototype should be minimized to less than \$1500. This is not an essential specification since more iterations of the board might be required to increase the performance of the

module. However, the cost of the module end-product should be less than \$500 is a non-essential specification because a low cost is not necessary for the design. The diagram on the case of the module is an essential specification because it will help the students to understand how the module functions. The size and weight of the module are both non-essential specifications because the module is intended to be used as a stationary device, and should not be moved when it is in-use. Finally, the compatibility of the software with computers that have less than 2 GB of RAM is an essential specification. This is because computers in university labs usually do not have the latest hardware, so compatibilities with lower end hardware must be provided to allow more students use the module.

Table 2. Non-functional specifications of the RF module

Specification	Description	Classification
Case	The module should be mounted in a protective case with no exposed parts except for the connectors. The protective case is designed in a way such that components are visible but not easily accessible.	Essential
Connectors	The input and output of each stage of the transmitter module, i.e. the voltage controlled oscillator, mixer, power amplifier, and bandpass filter are connected to an external 50 Ω SMA connector. A total of seven SMA connectors is used in the module.	Essential
Cost of development	The cost of the module prototype should be no more than \$1500.	Non-essential
Cost of prototype	The cost of the developed module should be no more than \$500.	Non-essential
Diagram	The casing of the module has block diagrams labelling each major stage of the transmitter module, including the voltage controlled oscillator, mixer, bandpass filters, and power amplifier.	Essential
Size of casing	The size of the casing not including the connectors should be no more than 40 cm x 10 cm x 8 cm.	Non-essential
Size of PCB	The size of the PCB should be small enough such that it can be contained in a casing that is 40 cm x 10 cm.	Non-essential
Software	The software used to control the module must be light-weight such that it works with systems that has less than 2 GB of RAM.	Essential
Weight	The weight of the module should be no more than 1.2 kg.	Non-essential

3. Risk Assessment

There are a few possible factors that can hinder the completion of the project. Table 3 lists these risks, their potential impact, their probability of occurrence and suggestions on how to prevent or reduce the effects of highly risky situations.

Table 3. Risks of not completing the project and possible solutions

Risk	Potential Impact	Probability	Mitigation
Insufficient knowledge – a programmable bandpass filter in the gigahertz range is a challenging design. The group members may not have enough knowledge to design a wide-range, low noise programmable filter.	Medium	High	To prevent this risk, the members can research relevant topics and seek consultation from professors. To reduce the impact of noisy filters, additional correcting designs (such as digital filters) can be added. If the range of the center frequency is not wide enough, an alternative of using multiple discrete filters can be adopted.
Technical – a USB interface may not be able to provide enough power for the entire transmitter.	Medium	Medium	
Insufficient time and skills - for RF circuits, in addition to usual design verifications, extra simulations and tests, such as electromagnetic interference, are needed. Because the group members have little prior experience with RF designs, it is likely that they need more time to learn and implement the full design and validation.	High	Medium	To reduce the likelihood of this risk, the members schedule to meet with the consultant weekly to ensure that the pre-set timelines can be met. The members can also consult RF professors on advice for RF design procedures or specific implementations. To mitigate the impact of this risk, the members can use the work term after the 4A term to catch up on RF design knowledge and work on computer simulations.

References

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- [2] Dream Catcher (2014, Nov. 27). *ME1000 Datasheet*. [Online]. Available:
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